MATH 372 Homework 5

①

$$A = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \qquad B = \begin{bmatrix} 5 & -3 & 0 \\ 6 & 1 & 0 \\ -5 & 2 & 2 \end{bmatrix}$$

$A_{3\times 1} \xrightarrow{\quad B \quad} \square_{3\times 3}$ Because of column of 1st matrics and row of 2nd matrics are different it cannot multiply.

$B_{3\times 3} \xrightarrow{\quad A \quad} \square_{3\times 1}$

$$B \cdot A = \begin{bmatrix} 5 & -3 & 0 \\ 6 & 1 & 0 \\ -5 & 2 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 5-6+0 \\ 6+2+0 \\ -5+4+0 \end{bmatrix} = \begin{bmatrix} -1 \\ 8 \\ -1 \end{bmatrix}$$

$$\boxed{B \cdot A = \begin{bmatrix} -1 \\ 8 \\ -1 \end{bmatrix}}$$

②

$$A = \begin{bmatrix} 3 & -2 & 0 \\ 5 & 1 & 1 \\ 2 & -2 & 0 \end{bmatrix} \rightarrow \left[\begin{array}{ccc|ccc} 3 & -2 & 0 & 1 & 0 & 0 \\ 5 & 1 & 1 & 0 & 1 & 0 \\ 2 & -2 & 0 & 0 & 0 & 1 \end{array}\right] \quad R_1 \leftrightarrow R_2$$

$$\left[\begin{array}{ccc|ccc} 5 & 1 & 1 & 0 & 1 & 0 \\ 3 & -2 & 0 & 1 & 0 & 0 \\ 2 & -2 & 0 & 0 & 0 & 1 \end{array}\right] \begin{array}{l} R_2 - \frac{3}{5}R_1 \rightarrow R_2 \\ R_3 - \frac{2}{5}R_1 \rightarrow R_3 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 5 & 1 & 1 & 0 & 1 & 0 \\ 0 & -\frac{13}{5} & -\frac{3}{5} & 1 & -\frac{3}{5} & 0 \\ 0 & -\frac{12}{5} & \frac{2}{5} & 0 & -\frac{2}{5} & 1 \end{array}\right] \quad R_3 - \frac{12}{13}R_2 \rightarrow R_3$$

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$$\begin{bmatrix} 5 & 1 & 1 & 0 & 1 & 0 \\ 0 & -13/5 & -3/5 & 1 & -3/5 & 0 \\ 0 & 0 & 2/13 & -12/13 & 2/13 & 1 \end{bmatrix} \xrightarrow{-\frac{13}{2} \cdot R_3 \to R_3}$$

$$\begin{bmatrix} 5 & 1 & 1 & 0 & 1 & 0 \\ 0 & -13/5 & -3/5 & 1 & -3/5 & 0 \\ 0 & 0 & 1 & -6 & 1 & 13/2 \end{bmatrix} \xrightarrow{R_2 + 3/5 R_3 \to R_2}$$

$$\begin{bmatrix} 5 & 1 & 1 & 0 & 1 & 0 \\ 0 & -13/5 & 0 & -13/5 & 0 & 39/10 \\ 0 & 0 & 1 & -6 & 1 & 13/2 \end{bmatrix} \xrightarrow{R_1 - R_3 \to R_1}$$

$$\begin{bmatrix} 5 & 1 & 0 & 6 & 0 & -13/2 \\ 0 & -13/5 & 0 & -13/5 & 0 & 39/10 \\ 0 & 0 & 1 & -6 & 1 & 13/2 \end{bmatrix} \xrightarrow{-\frac{5}{13} R_2 \to R_2}$$

$$\begin{bmatrix} 5 & 1 & 0 & 6 & 0 & -13/2 \\ 0 & 1 & 0 & 1 & 0 & -3/2 \\ 0 & 0 & 1 & -6 & 1 & 13/2 \end{bmatrix} \xrightarrow{R_1 - R_2 \to R_1}$$

$$\begin{bmatrix} 5 & 0 & 0 & 5 & 0 & -5 \\ 0 & 1 & 0 & 1 & 0 & -3/2 \\ 0 & 0 & 1 & -6 & 1 & 13/2 \end{bmatrix} \xrightarrow{1/5 \cdot R_1 \to R_1}$$

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 0 & -1 \\ 0 & 1 & 0 & 1 & 0 & -3/2 \\ 0 & 0 & 1 & -6 & 1 & 13/2 \end{bmatrix}$$

$$\therefore A^{-1} = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -3/2 \\ -6 & 1 & 13/2 \end{bmatrix}$$

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(3)

$$A = \begin{bmatrix} 5 & 7 & 4 \\ 3 & -1 & 3 \\ 6 & 7 & 5 \end{bmatrix} \qquad A^{-1} = \begin{bmatrix} 26 & 7 & -25 \\ -3 & -1 & 3 \\ -27 & -7 & 26 \end{bmatrix}$$

(a)

$$5x + 7y + 4z = 1$$
$$3x - y + 3z = 1$$
$$6x + 7y + 5z = 1$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 26 & 7 & -25 \\ -3 & -1 & 3 \\ -27 & -7 & 26 \end{bmatrix} \times \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 26+7-25 \\ -3-1+3 \\ -27-7+26 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ -1 \\ -8 \end{bmatrix}$$

$$\therefore (x, y, z) = (8, -1, 8)$$

(b) $5x + 7y + 4z = -1$

$3x - y + 3z = 2$

$6x + 7y + 5z = -1$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 26 & 7 & -25 \\ -3 & -1 & 3 \\ -27 & -7 & 26 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} -26+14+25 \\ 3-2-3 \\ 27-14-26 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 13 \\ -2 \\ -13 \end{bmatrix}$$

$$\therefore (x, y, z) = (13, -2, -13)$$

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(4) $\dfrac{7x-3}{x^3+2x^2-3x}$

$$\frac{7x-3}{x^3+2x^2-3x} = \frac{7x-3}{x(x^2+2x-3)} = \frac{7x-3}{x(x-1)(x+3)}$$

$$\Rightarrow \frac{7x-3}{x^3+2x^2-3x} = \frac{A}{x} + \frac{B}{(x-1)} + \frac{C}{(x+3)}$$

$$\Rightarrow A(x-1)(x+3) + B(x)(x+3) + C(x)(x-1) = 7x-3$$

$$\Rightarrow (Ax-A)(x+3) + Bx^2 + 3Bx + Cx^2 - Cx = 7x-3$$

$$\Rightarrow Ax^2 + Bx^2 + Cx^2 + 2Ax + 3Bx - Cx + 3A = 7x-3$$

$$\Rightarrow x^2(A+B+C) + x(2A+3B-C) + 3A = 7x-3$$

$$\Rightarrow \left.\begin{aligned} A+B+C &= 0 \\ 2A+3B-C &= 7 \\ 3A &= -3 \end{aligned}\right\} \quad \boxed{A = -1}$$

$$\Rightarrow \begin{aligned} A+B+C &= 0 \\ +\ 2A+3B-C &= 7 \\ \hline \boxed{3A+4B} &= 7 \end{aligned} \Rightarrow 3(-1)+4B = 7 \Rightarrow 4B = 10$$

$$\boxed{B = 5/2}$$

$$\Rightarrow A+B+C = 0 \Rightarrow -1 + 5/2 + C = 0 \Rightarrow C = 1 - 5/2$$

$$\Rightarrow \boxed{C = -3/2}$$

$$\therefore \boxed{\frac{7x-3}{x^3+2x^2-3x} = \frac{-1}{x} + \frac{5/2}{(x-1)} + \frac{-3/2}{(x+3)}}$$

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(5)

Let a = length, b = width

diameter $= 20''$

$a^2 + b^2 = 20^2 \Rightarrow a^2 + b^2 = 400$

area of rectangle $= 160 \Rightarrow a \times b = 160$

$\Rightarrow \boxed{b = \frac{160}{a}} \Rightarrow a^2 + \left(\frac{160}{a}\right)^2 = 400$

$\Rightarrow a^2 + \frac{25600}{a^2} = 400 \Rightarrow a^4 + 25600 = 400a^2$

$\Rightarrow a^4 - 400a^2 + 25600 = 0$

$x = \frac{400 \pm \sqrt{(-400)^2 - 4 \cdot (1)(25600)}}{2}$

$x^2 = \frac{400 \pm \sqrt{57600}}{2} \Rightarrow x^2 = \frac{400 \pm 240}{2}$

$x^2 = \frac{640}{2} \Rightarrow x^2 = 320 \Rightarrow x = \sqrt{320} \Rightarrow \boxed{x = 17.889}$

$x^2 = \frac{160}{2} \Rightarrow x^2 = 80 \Rightarrow x = \sqrt{80} \Rightarrow \boxed{x = 8.944}$

$\therefore a \times b = 160 \Rightarrow 17.889 \times 8.944 = 160$

$\therefore \boxed{a = 17.889}$, $\boxed{b = 8.944}$

The rectangle is 17.889 by 8.944 inches.

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(6)

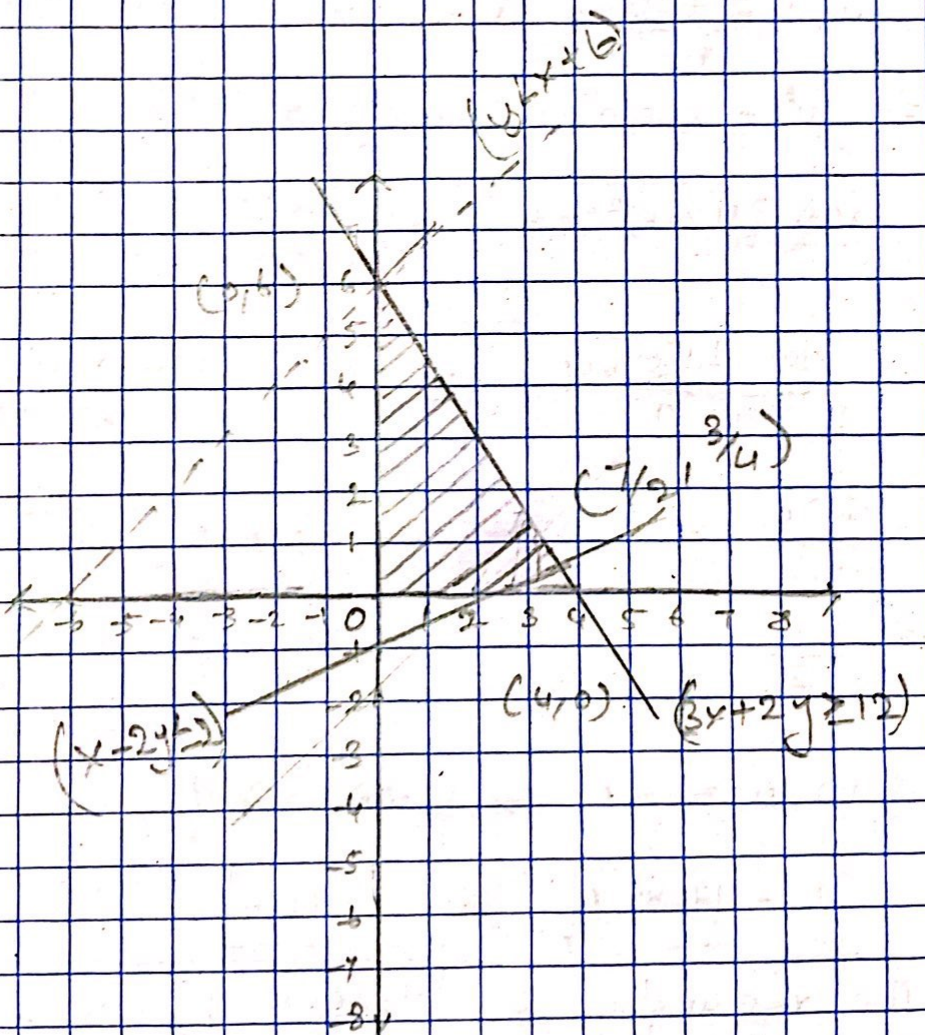
$$\left.\begin{array}{l} y < x + 6 \\ 3x + 2y \geq 12 \\ x - 2y \leq 2 \end{array}\right\}$$

$$\begin{array}{r} 3x + 2y = 12 \\ x - 2y = 2 \\ \hline 4x \quad = 14 \end{array} \Rightarrow \boxed{x = 7/2}$$

If $x = 7/2 \Rightarrow$ Then $\Rightarrow \frac{7}{2} - 2y = 2 \Rightarrow \frac{7}{2} - 2 = 2y$

$\Rightarrow \frac{3}{2} = 2y \Rightarrow \boxed{y = \frac{3}{4}}$

$\therefore (x, y) = (7/2, 3/4)$



(7)

$x - 7y^2 = 0 \Rightarrow x = 7y^2$

$\boxed{y^2 = 4px} \Rightarrow x = \left(\frac{1}{4p}\right) \cdot y^2$

From Comparing both equation

$\frac{1}{4p} = 7 \Rightarrow 4p = \frac{1}{7} \Rightarrow \boxed{p = \frac{1}{28}}$

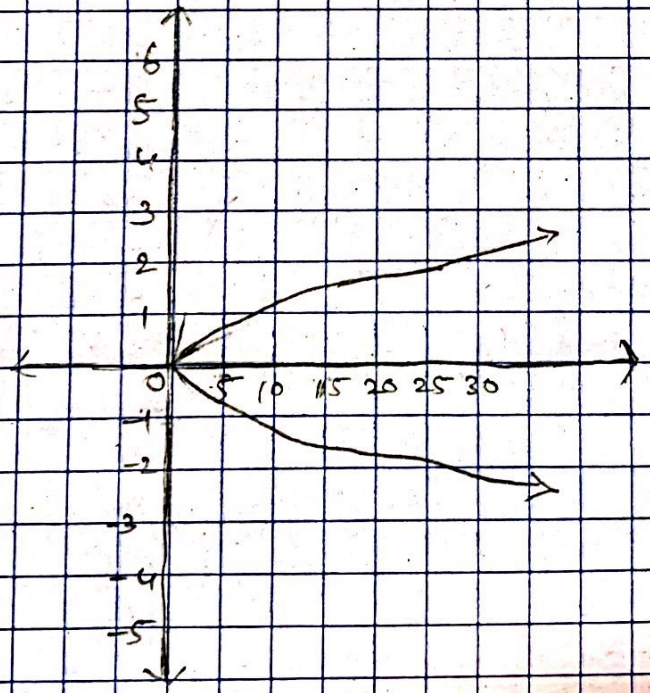
∴ Focus at : $\left(\frac{1}{28}, 0\right)$

vertex : $(0, 0)$

Directrix : $\boxed{x = -\frac{1}{28}}$

$\boxed{\text{Focal diameter} : \frac{1}{7}}$

x	y
0	0
7	1
7	-1
28	2
28	-2



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